Multimedia Search and Retrieval
Introduction

Winter Term 2025/26, 344.038, KV, 3h (4.5 ECTS)

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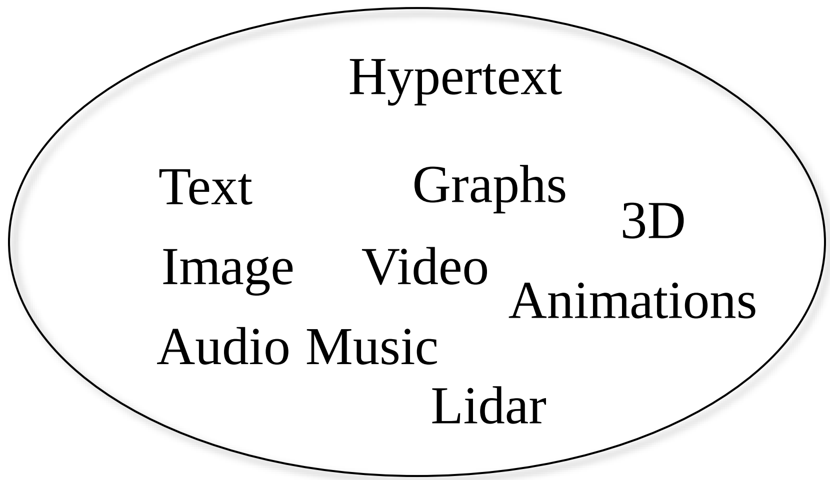
MULTIMEDIA MINING AND SEARCH GROUP

Quick Start: Introduction to MMSR



Zillions of digital media items

- immersion via Web, OSNs, streaming, etc.
- "democratization" of creation, production, and publishing
- constantly growing MM data

We need to develop strategies and tools for **managing and processing** this heterogeneous data, esp. for searching in abundant MM data

We need strategies to **describe** what "is in these data objects", i.e., what the data means, what it contains

We need structures to **represent the contents** (efficiently and effectively) of the media objects and the multi-faceted **relations** between them

Definitions: Multimedia

"Multimedia is essentially any digital data, including plain text, mostly unstructured, that we use to **communicate or capture information**. Beyond text, multimedia is visual and sound data, pictures, graphs, images, videos, animations, speech, music, sounds, and even 3D visualizations."

[Ponceleón and Slaney, 2011]

"Multimedia refers to a **collection of media types used together**. (Notice that a collection **may have only one member**.) At least **one of the media types must be non-alphanumeric**. […] We use the term multimedia object to refer to multimedia data to which a certain meaning has been attached."

[Blanken et al., 2007]

"The 3 modalities Audio, Visual, and Text are basic atoms describing various media items. A few examples of media items: picture (V), pop song (A + T), piano sonata (A), silent movie (V), regular movie (A + V + T). We consider all examples in which a single modality is used as **unimodal** or atomic media items; when more than one modality is included as **multimodal** or composite."

[Deldjoo et al., 2020]

Definitions: (MM)IR

"The IR Problem: the primary goal of an I(nformation)R(etrieval) system is to retrieve **all** the documents that are **relevant** to a user query while retrieving as few non-relevant as possible."

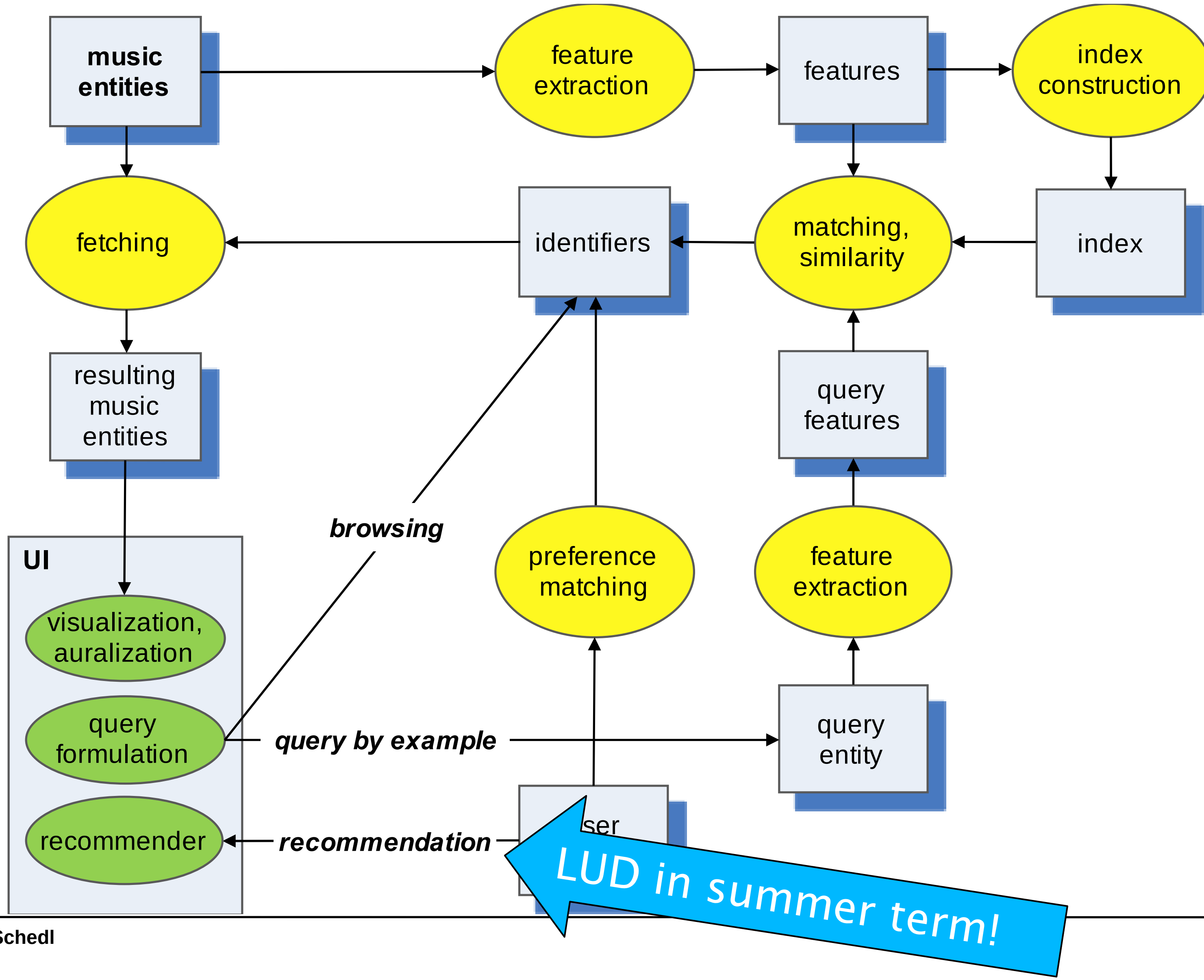
[Baeza-Yates and Ribeiro-Neto, 2011]

"The task of a Multimedia IR system is to retrieve text, image, video and sound data related to the interest of the user and rank them accordingly to a degree of relevance with regard to the user query. The **similarity** degree (i.e., the **ranking**) should be computed in order to improve the likelihood that the user will find the answers relevant. […]

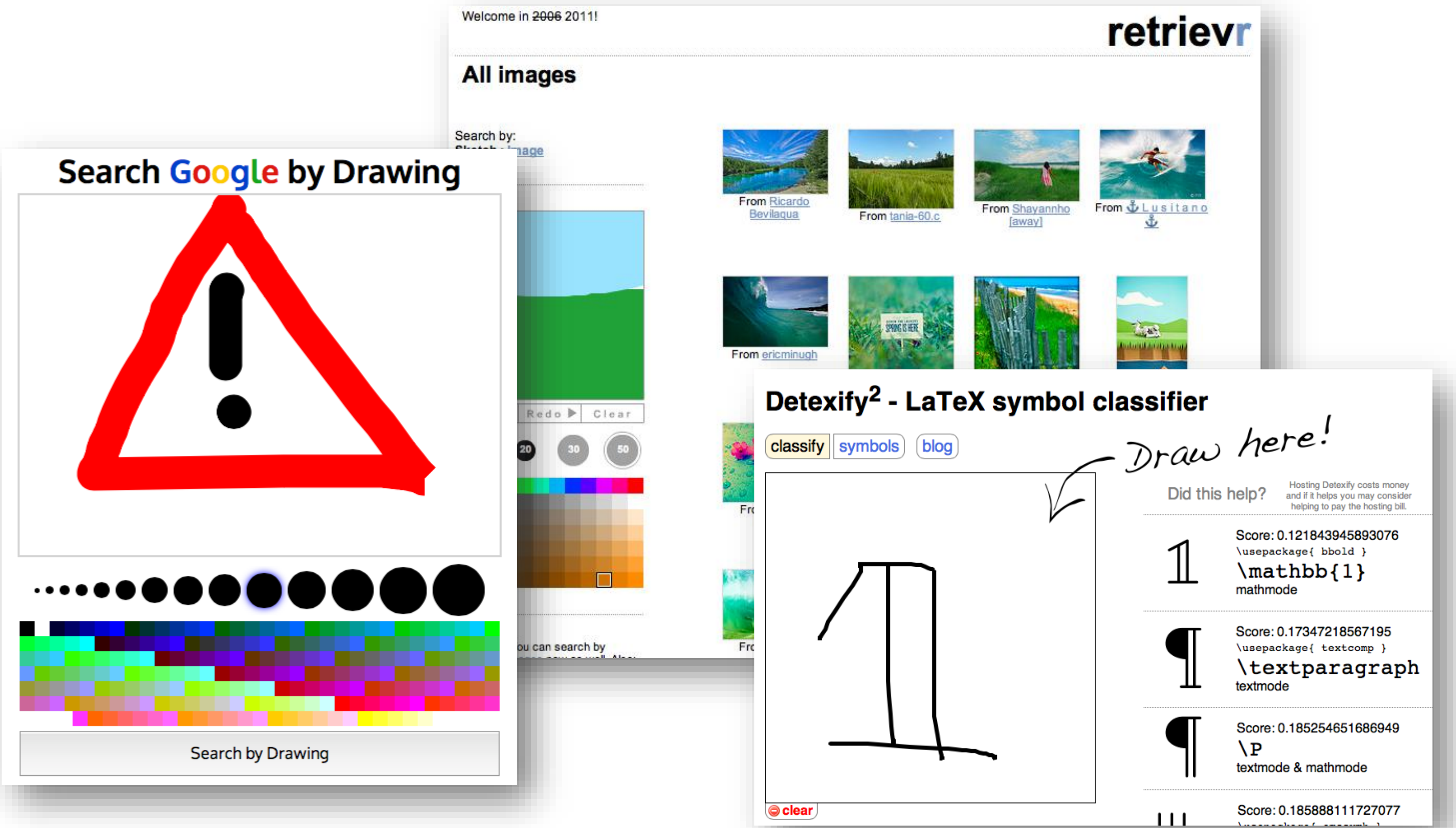
Multimedia information retrieval (MMIR) deals with searching, indexing, extracting, browsing and summarizing information and semantics contained in multimedia data."

[Ponceleón and Slaney, 2011]

Information Access Pipeline (Ex.: Music)



Examples: Query-by-Drawing



Text Data

Data signal: sequential stream of alphanumeric characters

Storage size: small

- 1 character = 1 byte (ASCII) – 4 byte (Unicode UTF-32)
- Book with 500 pages: approx. 2MB (consumption time 1 week?)

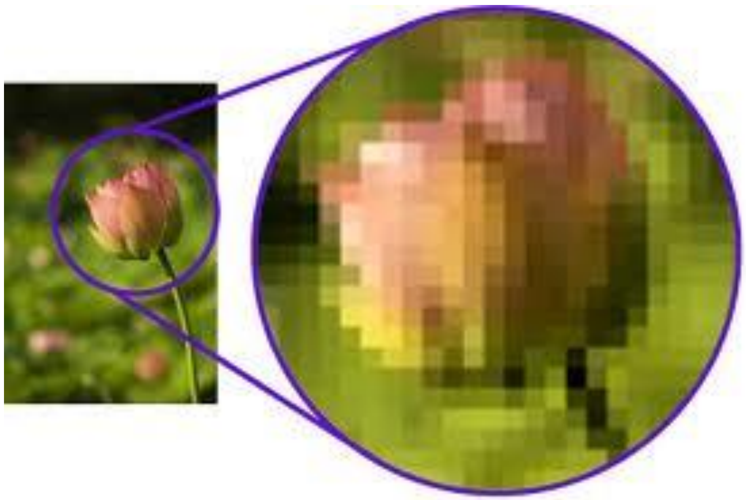
- Words are basic semantic units; redundant representation
- Semantic segmentation of a text data stream is easy due to punctuation, delimiters, and language grammar rules.
- Text forms with more structure and metadata (HTML, XML, JSON, etc.)
- Can be (easily) summarized, facilitates browsing

Image Data

Data signal: two-dimensional grid of color intensity values

Storage size: medium

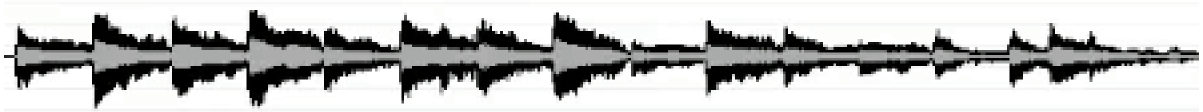
- Uncompressed color bitmap with 1920x1080 pixel: ~6MB
- 15 megapixel picture (JPEG compressed): ~3MB
- Consumption time: few seconds



- No structure inherent, segments have to be determined
- Summarization difficult (region of interest?)

Audio Data

Data signal: sequential stream of discrete amplitude values



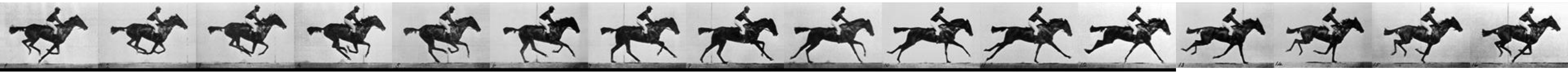
Storage size: medium

- 1 hour of CD audio: 635MB
- 1 hour of compressed audio: approx. 50MB
- Consumption time of an item (e.g., song): few minutes

- Semantic units need to be determined (difficult)
- Delimiters need to be determined; uninterrupted linear stream
- Summarization difficult (in music: chorus?)

Video Data

Data signal: stream of sequential images



Storage size: large

- 1 sec of uncompressed video: image size * fps
- 1 hr of video (8K Ultra-HD, 7680 x 4320px ~ 33Mpx ~ 100MB/frame!): approx. 10TB (uncompressed) or ~100 GB (compressed) per hour; highly redundant content between frames

- Spatial and temporal dimensions are important
- MPEG1/2 compression based on pixel differences, thresholding of difference allows *shot segmentation*
- MPEG4 supports *video objects* (e.g., a person), i.e., compression already provides “semantic” structuring

Metadata

Structured data describing (multi)media objects

- **Text**: e.g., Markup, HTML, XML, JSON, LaTeX
- **Audio**: e.g., recording/editing annotations
- **Music**: e.g., ID3 tags, MusicXML (for sheet music), Music Ontology, user-generated tags
- **Images**: e.g., EXIF data, user-generated tags
- **Video**: e.g., transcripts of spoken content, scene descriptions, user-generated tags

Searching for multimedia data is often searching in metadata!

Content-based features are sometimes considered metadata of the content, e.g., in MPEG7

Features and Similarity

- Digitizing media results in a complex and large data object
→ allows for compression & reconstruction (unnoticeable differences)
→ consumption/interpretation/etc. by humans
- For machine-based tasks, this representation is impractical (machine does not know a priori how to interpret raw data)
- We need to find a *numeric representation* that captures "important" properties of the data → **features**
- Typically more than one important property
→ multi-dimensional features → represented as a **vector**
→ multi-dimensional space
- "Similarity" of data items can be calculated as (inverse) distance of their *feature vectors*
- Retrieval: finding the most similar items to the query vector

Feature Space: 3-dim. Example

media collection

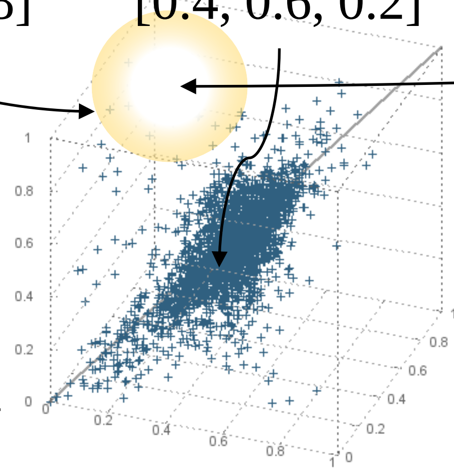
…

"query"

Feature Extraction

[0.0, 0.6, 0.8] [0.4, 0.6, 0.2] … [0.3, 0.7, 0.9]

3-dim Feature Space (redness, greenness, roundness)



Similarity … inverse distance in feature space

Retrieval … finding those items with highest similarity (smallest distance) to the query in feature space

From Low-level to High-level Features

High

- descriptions meaningful to the user
- semantic concepts *"how a human understands the data"*
- hard to extract (but has become "easier" thanks to DL)

Mid

- typically a (supervised) combination of low-level features to approx. high-level features
- aim to bridge the "semantic gap"

"Semantic Gap"

Low

- describe data patterns and statistics
- extracted automatically *"how the machine understands the data"*
- depend strongly on medium

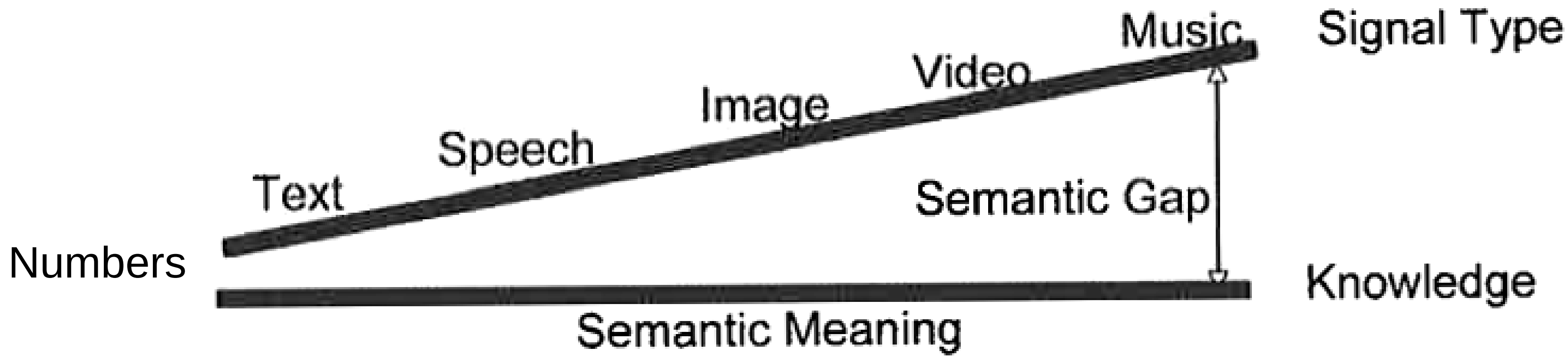
digital media representation (raw data)

From Low-level to High-level Features

	Text	Image	Video	Audio/Music
High	words, “meaning,” style, story	description, inherent context, aesthetics	event description, story, genre, interestingness, violence	structure, motifs, tempo, rhythm, chords, style/genre
Mid	words, concept clusters, sentiments	objects, ROIs	object motion, optical flow, speech extraction	MFCCs, rhythm patterns, “sound objects”
Low	characters, words, PoS	color, contrast, shape, texture	keyframe image/audio features, shot boundaries	energy, ZCR, spectral flux

digital media representation (raw data)

The Semantic Gap in Multimedia



- A strategy for overcoming the semantic gap is to map the problem to a representation with smaller semantic gap
- E.g., Video → speech extraction → Speech → speech recognition → Text → semantic modeling → Numbers ✔

Agenda

- 08.10.2025: Introduction to Multimedia Retrieval
- 15.10.2025: Text Features I
- 22.10.2025: Text Features II and Evaluation of IR Systems (Only via Zoom!)
- 29.10.2025: *No meeting!*
- 05.11.2025: Introduction to Exercise
- 12.11.2025: Audio and Music Features I
- 19.11.2025: Audio and Music Features II
- 26.11.2025: Image Features
- 03.12.2025: Video Features and Multimodal Data Fusion
- 10.12.2025: Modern Multimodal Models (guest lecture by Shah Nawaz)
- 17.12.2025: Lighting Talks and Discussion
- 07.01.2026: *No meeting!*
- 14.01.2026: Lighting Talks and Discussion
- 21.01.2026: Written Exam (via Zoom+Moodle)
- 28.01.2026: Students’ Final Presentation of Results

Practical Part

- Objective: Create a multimodal music retrieval system
- Consists of various tasks:
  - Basics: working with Music4All-Onion dataset, devising and implementing baseline IR systems, creating a web-based UI for retrieval, perform qualitative analysis of IR results
  - Implement and integrate IR systems that are based on unimodal similarity
  - Integrate hybrid ranking models based on additional modalities
  - Train, use, and evaluate a NN-based IR system
  - Experiment with different relevance definitions
  - Conduct structured quantitative evaluation experiments assessing all systems w.r.t. accuracy and beyond-accuracy metrics

Practical Part

- Objective: Create a multimodal music retrieval system
- In groups of 5 students (groups formed by students or optionally, random assignment)
  - If you want (and are able) to form a group yourself, write an email with all group members to markus.schedl@jku.at by **October 12 (23:59)**
  - All remaining students (who opted-in) will be assigned randomly to additional (new) groups!
- Accounts for 60% of your final grade!
- For Q&A you should use the Moodle Forum
- Final report and (optional) intermediate report

Success Criteria

- **Practical task** (max. 60 pts)
  - **Report (45 pts):**
    - **Presentation (10 pts):** structure, readability, clarity, comprehensibility: *scientific style and depth* (e.g., clear and detailed description of methodology)
    - **Approach (10 pts):** technical soundness and justification of your algorithmic choices, reproducibility (code!)
    - **Experiments (10 pts):** suitedness and soundness of experimental setup, clarity, following good scientific practice
    - **Analysis and Discussion of Results (15 pts):** comprehensibility (e.g., how results are communicated to the reader), depth of analysis and discussion, conclusion
  - **Presentation (7 pts)**: quality of presentation (approach, results, analysis, and demonstration of IR system) and discussion (Q&A)
  - **Final IR system/UI (8 pts):** usability
- **Written exam** (max. 40 pts)

Both lecture and exercise parts need to be positive to get a positive grade! (>50% on both parts)

Now it's time to opt-in (or out)

Students registered in KUSSS and opting-in in the first meeting will be graded! If you cannot attend the first meeting and still want to participate, email me as soon as possible.